East West University
Department of CSE

Course Code: **CSE 347**
Course Title: **Information System Analysis and Design**

Project: **StudyByte – Peer-to-Peer Micro-Tutoring**

Task: **CPM & PERT Analysis**

Semester and Year:
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GROUP NO: **04**

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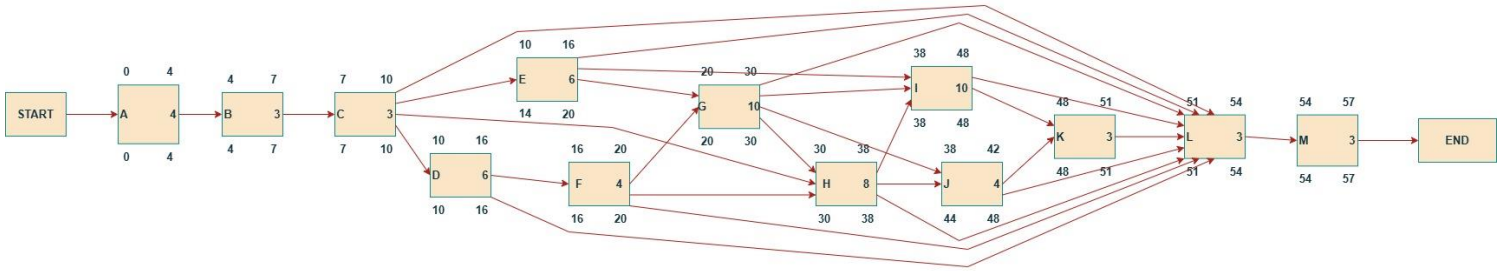
Part 1: Critical Path Method (CPM)

1.1 CPM Network Diagram



1.1 CPM Table

ES = Early Start | **EF** = Early Finish
LS = Late Start | **LF** = Late Finish
TF = Total Float | **FF** = Free Float
CP = Critical Path

Activity	Description	Dependency	Duration
A	Feasibility Analysis	—	4
B	Requirement Gathering	A	3
C	Requirement Analysis	B	3
D	System Design	C	6
E	Database Design	C	6
F	UI/UX	D	4
G	Implementing User Module Features	E, F	10
H	Implementing Booking Module	F, G, C	8
I	Implementing Wallet & Escrow	E, G, H	10
J	Implementing Rating Feature	G, H	4
K	Implementing Admin Control	I, J	3
L	Testing	C, D, E, F, G, H, I, J, K	3

Activity	Description	Dependency	Duration
M	Documentation	L	3

1.2 CPM Total Float & Free Float

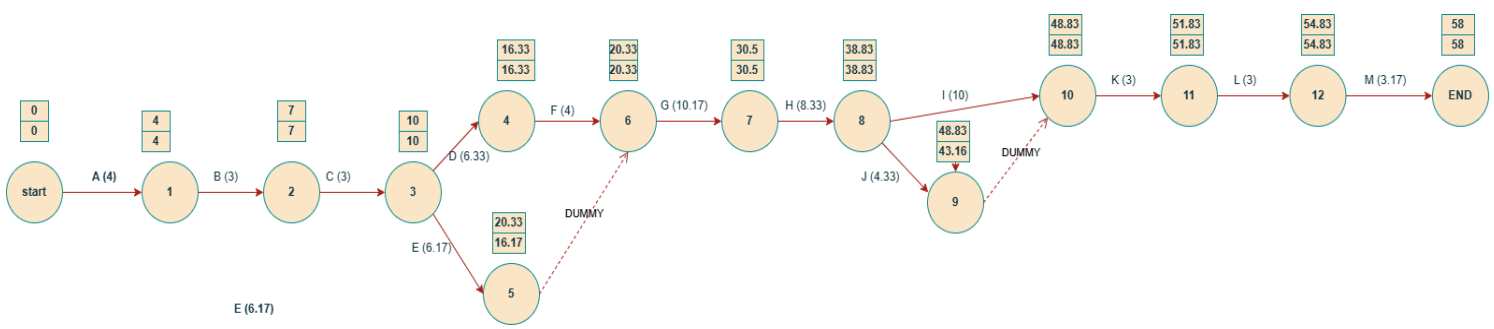
Activity	ES	EF	LS	LF	Total Float	Free Float	CP
A	0	4	0	4	0	0	✓
B	4	7	4	7	0	0	✓
C	7	10	7	10	0	0	✓
D	10	16	10	16	0	0	✓
E	10	16	14	20	4	4	—
F	16	20	16	20	0	0	✓
G	20	30	20	30	0	0	✓
H	30	38	30	38	0	0	✓
I	38	48	38	48	0	0	✓
J	38	42	44	48	6	6	—
K	48	51	48	51	0	0	✓
L	51	54	51	54	0	0	✓
M	54	57	54	57	0	0	✓

1.2 CPM Summary

Metric	Value
Project Length (Duration)	57 Days
Critical Path	A → B → C → D → F → G → H → I → K → L → M
Number of Critical Activities	11 out of 13
Non-Critical Activities	E, J

Part 2: Program Evaluation & Review Technique (PERT)

2.1 PERT Network Diagram



2.2 PERT Table

O = Optimistic | **M** = Most Likely (original) | **P** = Pessimistic

te = Expected Time = (O + 4M + P) / 6

σ² = Variance = ((P−O)/6)²

Activity	ID	Dependency	O	M	P	te	σ (Std Dev)
Feasibility Analysis	A	—	2	4	6	4.00	0.67

Activity	ID	Dependency	O	M	P	te	σ (Std Dev)
Requirement Gathering	B	A	1	3	5	3.00	0.67
Requirement Analysis	C	B	1	3	5	3.00	0.67
System Design	D	C	4	6	10	6.33	1.00
Database Design	E	C	4	6	9	6.17	0.83
UI/UX	F	D	2	4	6	4.00	0.67
Implementing User Module Features	G	E, F	7	10	14	10.17	1.17
Implementing Booking Module	H	F, G, C	6	8	12	8.33	1.00
Implementing Wallet & Escrow	I	E, G, H	6	10	14	10.00	1.33
Implementing Rating Feature	J	G, H	2	4	8	4.33	1.00
Implementing Admin Control	K	I, J	2	3	4	3.00	0.33
Testing	L	C, D, E, F, G, H, I, J, K	1	3	5	3.00	0.67
Documentation	M	L	2	3	5	3.17	0.50

2.2 PERT Summary

Metric	Value
PERT Critical Path	A → B → C → D → F → G → H → I → K → L → M
Expected Project Length (ΣTE)	58 Days
Sum of Variances (σ^2) on Critical Path	8.20 days^2

Metric	Value
Standard Deviation (σ)	2.86 days